EXCEL PIVOT TABLES

Complete Learning & Practical Reference Handbook

Lessons 16-27 | From first Pivot Table to interactive business dashboard

Explanations • Excel navigation • Field configurations • Sample datasets • Expected outputs • Best practices

Includes all instructional concepts and worked examples from the lesson series. Practice assignments and quizzes have been intentionally omitted.

Contents and how to use this handbook

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16	Interface, first Pivot Table, Rows / Columns / Values / Filters
17	Value Field Settings; SUM / COUNT / AVERAGE / MAX / MIN; source expansion
18	Agent-wise analysis, multi-level rows, columns, totals and layout
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20	Currency and number formatting; labels; preserve formatting
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Use desktop Excel on Windows. Menu wording may vary slightly by version. All example data and outputs below are illustrative practice examples from our learning series; figures from the lecturer’s larger source workbook will differ.

LESSON 16 | Pivot Table fundamentals and interface

Goal: create a Pivot Table, recognize its two-part interface, and use each of the four field areas to answer a business question.

Source data and why analysis is separate

A Pivot Table summarizes many source records without changing them. Excel normally suggests placing the analysis in a new worksheet, keeping original transactions distinct from the report. It creates a report area and a PivotTable Fields pane that lists the headers of the chosen source. Pivot Tables use cached source data; source edits usually require Refresh.

Order ID	Product	Final Price	Agent	Order Type
O101	Pizza	750	Adrian	Online
O102	Burger	500	Albion	Physical Visit
O103	Pizza	850	Roche	Online
O104	Sandwich	300	Adrian	Physical Visit
O105	Burger	600	Albion	Online
O106	Pizza	900	Roche	Physical Visit
O107	Sandwich	350	Adrian	Online
O108	Burger	550	Albion	Physical Visit
O109	Pizza	800	Roche	Online
O110	Sandwich	400	Adrian	Online

Insert the first Pivot Table

1. Select the full source dataset A1:E11 (including headers).

2. Choose Insert > PivotTable > From Table/Range.

3. Confirm the table/range; select New Worksheet; click OK. Rename the analysis sheet 16_FirstPivot.

4. In PivotTable Fields, drag Final Price to Values. Excel generally creates Sum of Final Price = 6000.

The four areas of the Fields pane

Area	Job	Example
Rows	Show named categories vertically	Product Name
Columns	Compare categories side by side	Order Type
Values	Calculate numbers	Sum of Final Price
Filters	Restrict the whole report	Agent Handled = Adrian

Worked output: product sales

Keep Final Price in Values; drag Product Name into Rows.

Product	Sales
Burger	1650
Pizza	3300
Sandwich	1050
Grand Total	6000

Worked output: product by channel

Add Order Type to Columns. The same 6000 is now summarized across products and order channels.

Product	Online	Physical Visit	Total
Burger	600	1050	1650
Pizza	2400	900	3300
Sandwich	750	300	1050
Grand Total	3750	2250	6000

Add Agent Handled to Filters, choose Adrian. Only Adrian’s source records contribute: Pizza 750; Sandwich 1050; overall 1800. Blank cross-category cells mean that no matching transaction exists.

> Remember: Rows = categories; Columns = comparisons; Values = calculation; Filters = records included. The order and placement of fields change the view, not the underlying source.

LESSON 17 | Value Field Settings and changing the source

Goal: perform multiple calculations on one numerical field and recover from a Pivot Table created from an incomplete source.

Why select the whole dataset?

Selecting only C1:C11 (Final Price) creates a Pivot Table whose Fields pane contains only Final Price. It can calculate a total but cannot slice results by Agent or Product. To repair it, click the Pivot Table > PivotTable Analyze > Change Data Source; replace the one-column reference with the full source, e.g. 16_PivotData!$A$1:$E$11; click OK. The other fields become available.

Default SUM and COUNT behavior

When a fully numeric field is dragged to Values, Excel usually selects SUM. A text field, such as Order Type, defaults to COUNT. If a numeric column contains blanks or text, COUNT may be chosen instead. COUNT counts nonblank field entries and is not a distinct count.

Using the same field more than once

1. In the PivotTable Fields pane, drag Final Price into Values five times. Excel creates separate entries such as Sum of Final Price, Sum of Final Price2, etc.

2. Open Value Field Settings for each copy. Choose Count, Sum, Average, Max and Min respectively.

3. In Custom Name, use Number of Sales, Overall Sales, Average Sales, Highest Sale and Lowest Sale.

4. Move the Values entries up or down to control their displayed order.

Measure	Calculation	Expected output
Number of Sales	Count of Final Price	10
Overall Sales	Sum of Final Price	6000
Average Sales	Average of Final Price	600
Highest Sale	Max of Final Price	900
Lowest Sale	Min of Final Price	300

Three routes to Value Field Settings

Method	Excel navigation
Ribbon	Select result cell > PivotTable Analyze > Field Settings
Context menu	Right-click result cell > Value Field Settings
Fields pane	Dropdown beside the item in Values > Value Field Settings

The dialog also offers Product, Count Numbers, standard deviation, and variance calculations. Value Field Settings controls aggregation; its Number Format button controls presentation.

> A custom Values caption must not exactly duplicate an existing source field name. A name such as Overall Sales is clearer than Sum of Final Price.

LESSON 18 | Agent-wise sales; Rows, Columns and hierarchy

Goal: report transaction count, overall sales and average sales for each agent, then compare Online and Physical Visit performance.

Create the agent summary

1. Insert a Pivot Table from the full 16_PivotData source (or copy an existing compatible Pivot Table).

2. Add Final Price to Values three times; configure Count, Sum and Average.

3. Drag Agent Handled into Rows. Use clear headings and currency format for the money fields.

Agent	Number of Sales	Overall Sales	Average Sales
Adrian	4	1800	450.00
Albion	3	1650	550.00
Roche	3	2550	850.00
Grand Total	10	6000	600.00

The grand-total average is 6000 / 10 = 600, not the unweighted average of the three agent averages (616.67). Pivot Table averages are calculated from source records.

Add a second Rows field

Copy the entire Pivot Table using PivotTable Analyze > Select > Entire PivotTable; Ctrl+C; paste into an empty analysis area. Drag Order Type underneath Agent Handled in Rows. This creates an Agent > Order Type hierarchy. Each agent's online and physical orders appear underneath their agent group.

Agent / Channel	Count	Sum	Average
Adrian - Online	3	1500	500
Adrian - Physical	1	300	300
Albion - Online	1	600	600
Albion - Physical	2	1050	525
Roche - Online	2	1650	825
Roche - Physical	1	900	900

Reverse the hierarchy or move a field into Columns

Drag Order Type above Agent Handled in Rows: the main groups become Online and Physical Visit, with agents beneath. Or use Rows = Order Type and Columns = Agent Handled to compare agents horizontally.

Channel	Adrian	Albion	Roche	Total sales
Online	1500	600	1650	3750
Physical Visit	300	1050	900	2250

Channel	Adrian	Albion	Roche	Total sales
Grand Total	1800	1650	2550	6000

Subtotals, grand totals and blank rows

PivotTable Design > Subtotals: choose not to show, or show at the top/bottom of groups. PivotTable Design > Grand Totals: On/Off for Rows and Columns. PivotTable Design > Blank Rows > Insert Blank Line After Each Item adds breathing room between groups. Rename Row Labels or Column Labels only where the label is editable; generic headings should communicate Agents and Mode of Purchase.

> A Pivot Table copied as a whole remains an editable Pivot Table; copying only displayed values creates a static output. Leave room for a Pivot Table to expand.

LESSON 19 | Sort, Label Filters, Value Filters and Report Filters

Goal: rank products by total revenue, select product names containing Crispy, and analyse online-only sales.

Illustrative product transactions

In our 15-order practice dataset the product totals are: Crispy Pizza 9400; Cheese Pizza 7600; Crispy Burger 6500; Paneer Pizza 6000; Veg Burger 4200. Total revenue = 33700.

Sort by a calculated measure

1. Create a Pivot Table from the full source. Put Product Name in Rows and Final Price (Sum) in Values.

2. Open the row field dropdown > More Sort Options. Choose Descending by Sum of Final Price / Overall Sales, not simple Z-to-A sorting of names.

3. Alternatively right-click an Overall Sales amount > Sort > Largest to Smallest.

Product	Overall Sales (descending)
Crispy Pizza	9400
Cheese Pizza	7600
Crispy Burger	6500
Paneer Pizza	6000
Veg Burger	4200

Label filter versus value filter

Filter	Meaning	How
Label Filter	Select categories based on names	Product dropdown > Label Filters > Contains > Crispy
Value Filter	Select categories based on summarized numbers	Product dropdown > Value Filters > Greater Than > Overall Sales > 6000
Report Filter	Select which source rows feed the whole report	Order Type to Filters > Online

A Label Filter of Contains "Crispy" leaves Crispy Pizza (9400) and Crispy Burger (6500). A Value Filter of Overall Sales > 6000 leaves Crispy Pizza, Cheese Pizza and Crispy Burger; Paneer Pizza at exactly 6000 is excluded. The Greater Than or Equal To choice would include it.

Why Product Name belongs in Rows for per-product questions

Placing Product Name only in Filters and selecting both Crispy products gives a combined sales total of 15900, but does not list each product separately. For "sales for each Crispy product," keep Product Name in Rows and filter its labels.

Online-only overall and average sales

1. Create a fresh Pivot Table or clear previous filters. Put Product Name in Rows.

2. Add Final Price twice in Values, configuring Sum = Overall Sales and Average = Average Sales.

3. Put Order Type in Filters, choose Online.

Product	Online sales	Average online sale
Cheese Pizza	5400	2700
Crispy Burger	3800	1900
Crispy Pizza	4000	2000
Paneer Pizza	3200	3200
Veg Burger	1400	1400

Only the eight Online transactions contribute to these results; online revenue = 17800. You may combine Report Filters with row-level Label Filters.

LESSON 20 | Number formatting, captions and report presentation

Goal: make a correct Pivot Table readable and keep the formatting consistent through updates.

Normal cell formatting vs Value Field Number Format

Home > Number formats only the selected cells. If you omit a row or new report values appear after refresh, the output may be inconsistent. Preferred method: right-click a Pivot Table result > Value Field Settings > Number Format, then select Currency, Number or Percentage and the desired decimal places. This format belongs to the entire Values field.

Measure	Recommended presentation	Settings
Overall Sales	7,600	Currency, 0 decimals
Average Sales	2,533.33	Currency, 2 decimals
Average Rating / CSAT	8.5	Number, 1 decimal
Number of Orders	15	Number, 0 decimals
Average Discount	8.33%	Percentage, 2 decimals

Rename fields and labels

1. Open Value Field Settings > Custom Name; rename Sum of Final Price to Overall Sales, Average of Final Price to Average Sales and Average of Rating Given to CSAT Score.

2. Where the report uses generic "Row Labels" or "Column Labels", replace their captions with meaningful labels such as Products or Mode of Purchase.

3. A displayed item like Physical Visit can sometimes be renamed by typing Offline directly over its label; this affects the displayed Pivot item, not source records, and may be shared by other Pivot Tables.

4. Place the report title outside the Pivot Table and adjust column widths (Home > Format > AutoFit Column Width). Keep numbers consistently aligned.

Preserve appearance on refresh

Right-click inside the Pivot Table > PivotTable Options > Layout & Format. Check Preserve cell formatting on update. Uncheck Autofit column widths on update when the dashboard uses carefully set widths. Repeat this setting for each Pivot Table; it is not global.

Products	Overall Sales	Average Sales	CSAT
Cheese Pizza	7600	2533.33	6.0
Crispy Burger	6500	2166.67	9.0
Crispy Pizza	9400	2350.00	7.8
Paneer Pizza	6000	3000.00	7.0
Veg Burger	4200	1400.00	7.0

Number formatting changes display precision, not underlying numerical calculations. Do not format customer ratings as currency or a percentage unless the metric truly calls for it.

LESSON 21 | Date analysis, grouping and refreshing

Goal: calculate revenue by day, understand automatic date grouping, correct a source date and update the report.

Daily sales

Create a Pivot Table from a dataset containing Contact Date, Final Price and Rating Given. Put Contact Date in Rows and Final Price (Sum) in Values. If Excel automatically groups dates into Years / Quarters / Months, remove unwanted fields or right-click an actual date > Ungroup to show daily records.

Date	Daily Sales
01-Nov-2023	3500
02-Nov-2023	3000
03-Nov-2023	5500
04-Nov-2023	3600
05-Nov-2023	6300
06-Nov-2023	5100
07-Nov-2023	4600
08-Nov-2023	4100
09-Nov-2023	3700
10-Nov-2023	6200
23-Dec-2023 (before correction)	2500

Correct a date and Refresh

1. In the source, change an incorrect Contact Date from 23-Dec-2023 to 23-Nov-2023.

2. Return to the Pivot Table: old date data may remain in the report because Pivot Tables use a cache.

3. Select the Pivot Table > PivotTable Analyze > Refresh; the 23-Nov row appears while 23-Dec disappears from the report. Overall revenue is unchanged.

4. Data > Refresh All refreshes applicable workbook data sources and reports; it is the broader command for multi-report workbooks.

Add customer satisfaction for each day

Add Rating Given into Values; Value Field Settings > Average; rename it CSAT Score; Number Format = Number, one decimal. For 01-Nov, ratings 8 and 7 give 7.5. A rating average is used here as the course's CSAT score; some organizations define CSAT differently.

Group dates into months, quarters and years

1. Click an actual date label, not a value result. Choose PivotTable Analyze > Group Field or right-click > Group.

2. Select Months, Quarters and/or Years as needed; omit hours/minutes/seconds when source records do not contain useful time detail.

3. For a day-by-quarter hierarchy, retain the quarter grouping and individual date field in Rows.

4. For datasets covering more than one year, group Months together with Years to avoid combining the same month across years.

Calendar quarter	Months
Q1	Jan-Mar
Q2	Apr-Jun
Q3	Jul-Sep
Q4	Oct-Dec

Use +/- expand/collapse buttons to show or hide detail. If grouping fails, check for text dates, invalid values or blanks. Old field items in dropdowns can sometimes persist because PivotTable Options > Data retains deleted items; set retention to None and refresh when appropriate.

LESSON 22 | Top and Bottom product rankings

Goal: show the three highest- and three lowest-selling products, with their average ratings.

Set up the product report

Using the six-product, 18-order practice dataset, put Product Name in Rows, Final Price (Sum) and Rating Given (Average) in Values. Name the measures Overall Sales and CSAT Score. This dataset adds a Sandwich category with 3000 sales, giving total revenue 36700.

Product	Sales	CSAT
Crispy Pizza	9400	8.5
Cheese Pizza	7600	6.7
Crispy Burger	6500	8.7
Paneer Pizza	6000	8.0
Veg Burger	4200	6.3
Sandwich	3000	5.3

Apply Top 3 by overall sales

1. Open the Product Name dropdown > Value Filters > Top 10.

2. Set Top > 3 > Items > By Overall Sales.

3. Click OK. The report shows Crispy Pizza (9400), Cheese Pizza (7600), Crispy Burger (6500), each with its CSAT.

Bottom 3 in a second Pivot Table

1. Copy the entire Pivot Table and paste it in another empty area, with space for potential expansion.

2. On the copied Pivot Table, open Product Name dropdown > Value Filters > Top 10.

3. Choose Bottom > 3 > Items > By Overall Sales.

4. Sort ascending by Overall Sales if you want the lowest product first.

Top 3	Sales	Bottom 3	Sales
Crispy Pizza	9400	Sandwich	3000
Cheese Pizza	7600	Veg Burger	4200
Crispy Burger	6500	Paneer Pizza	6000

Verify rankings refresh dynamically

In the source data, increase a Paneer Pizza sale from 2800 to 7800. Paneer Pizza total becomes 11000, so after PivotTable Analyze > Refresh it enters the Top 3. Return the original price and refresh again. A saved Top 3 Value Filter re-evaluates the ranking; a manually chosen list of product names does not. More Sort Options controls order but does not itself filter out categories.

Do not confuse Top 3 by Overall Sales with Top 3 by CSAT. The ranking measure is chosen in the Top 10 dialog. Tied values may affect the exact number of shown items.

LESSON 23 | Fixed ranges, changing data source and Excel Tables

Goal: ensure a Pivot Table includes transactions added after the report is created.

The fixed-range problem

A Pivot Table created from 23_SourceData!$A$1:$E$11 knows only those 10 rows of transactions plus headers. When a new Pizza order (1200) is added in row 12, refreshing the Pivot Table still reports 6000 total because row 12 falls outside its original source.

Manually expand the Pivot Table source

1. Select the Pivot Table > PivotTable Analyze > Change Data Source.

2. Change the source from 23_SourceData!$A$1:$E$11 to 23_SourceData!$A$1:$E$12.

3. Confirm, refresh if needed; total becomes 7200, and Pizza sales increase from 3300 to 4500.

Whole columns versus an Excel Table

Referencing A:E can capture additional rows within those columns, but may introduce blank items and unnecessary processing. A structured Excel Table is normally preferable for regularly updated business data. Whole-column sources can expose (blank) category items, which you may hide through the field dropdown.

Convert the source to a named Table

1. Select the clean data range, including headers; press Ctrl+T or Home > Format as Table.

2. Confirm the range and check My table has headers.

3. Click within the Table > Table Design > Table Name. Enter SalesData (without spaces).

4. Add new records directly beneath the Table; Excel usually extends the Table automatically. Adding a header directly to the right can similarly extend its columns.

5. Back in the Pivot Table, choose PivotTable Analyze > Change Data Source; use SalesData as the source; confirm.

Expansion worked example

Stage	Added record	Expected sales total
Original	10 orders	6000
Expand fixed range	O111 Pizza 1200	7200
After Table expansion	O112 Burger 1000; O113 Pizza 1500	9700
Refresh after another Table row	O114 Sandwich 800	10500

After each addition to the Excel Table, refresh the Pivot Table. An expanding Table and refreshing Pivot Table are different operations: the Table range may expand immediately, while the cached report remains stale until Refresh.

LESSON 24 | Grouping numerical ratings, weeks and product text

Goal: create category-level analysis without editing the original source schema.

Group customer ratings by selected values

1. Create a Pivot Table from 24_GroupingData and place Rating Given in Rows.

2. Ctrl-click rating labels 2, 3 and 4; right-click > Group. Rename Group1 to Poor Satisfaction.

3. Ctrl-click 5, 6 and 7; Group; rename Neutral Satisfaction.

4. Ctrl-click 8, 9 and 10; Group; rename High Satisfaction.

5. Keep the created grouped field (for example Rating Given2) in Rows; remove the original Rating Given field from Rows when you want a one-row-per-group summary.

Grouped category	Ratings used	Sales	Count of Customer ID
Poor Satisfaction	2-4	7900	4
Neutral Satisfaction	5-7	14100	6
High Satisfaction	8-10	13800	6
Grand Total		35800	16

To calculate these values, put Final Price (Sum) and Customer ID (Count) into Values. Count of Customer ID is not automatically a distinct customer count; repeated customers with multiple orders are counted multiple times.

Group dates into 7-day intervals

1. Create a date-based Pivot Table, add Contact Date to Rows, then select an individual date label.

2. Choose PivotTable Analyze > Group Field or right-click > Group.

3. Deselect other intervals, select Days, enter Number of days = 7, and set Starting at = 01-Nov-2023.

4. Retain the grouped week field in Rows and add Final Price (Sum) to Values. Rename group captions Week 1 through Week 5 if useful.

Group	Period	Sales
Week 1	01-07 Nov	9200
Week 2	08-14 Nov	6600
Week 3	15-21 Nov	11400
Week 4	22-28 Nov	6500
Week 5	29 Nov-05 Dec	2100

These are sequential seven-day intervals from the Starting at date, not necessarily Monday-to-Sunday weeks. The last group extends into December even if only November transactions are present. To show daily detail within weeks, include the original date under the grouped field in Rows.

Manually group text products

1. Put Product Name in Rows. Ctrl-select Crispy Pizza and Medium Crispy Burger > Group > rename Crispy.

2. Ctrl-select Paneer Tikka Pizza and an added example product Spicy Tikka Burger > Group > rename Spicy.

3. Ctrl-select Cheese Pizza, Minty Pizza and Veg Burger > Group > rename Others.

4. Keep the grouped product-category field in Rows. Add Final Price (Sum), Rating Given (Average) and Order ID (Count) to Values; optionally add Order Type beneath category in Rows to compare channels.

> Adding Spicy Tikka Burger (1800, rating 8) changes the 16-row grouping dataset to 17 records. The course practice values shown earlier for this final text-grouping section were not internally consistent; use your Pivot Table results from the actual updated source as authoritative rather than copying an inconsistent total.

Shared grouping cache caution

Grouping can affect other Pivot Tables sharing the same Pivot Table cache. You do not need to delete your other reports automatically. For stable business categories consider adding a Product Category column to the source, or use a separate cache/source for independent grouping.

LESSON 25 | Slicers and Report Connections

Goal: filter multiple business summaries with interactive category buttons.

Create the interactive reports

Use the 15-order practice data with Agent Handled, Order Type, Product Name, Final Price and Rating Given. Create an Overall Sales Pivot Table, then copy the entire Pivot Table and alter copies for agent-wise and product-wise sales. Copying helps preserve compatibility for Report Connections.

Insert Order Type Slicer

1. Click inside an existing Pivot Table > PivotTable Analyze > Insert Slicer.

2. Select Order Type and click OK. A floating object shows Online and Physical Visit buttons.

3. Click Online to filter its connected report, or Physical Visit to show that channel.

4. Select the Slicer border > Slicer ribbon > Report Connections; check all compatible Pivot Tables and click OK.

5. Test again. All connected Pivot Tables should change together.

Selected channel	Overall Sales
No filter	33700
Online	17800
Physical Visit	15900

Add Agent and Rating slicers

Insert a second Slicer for Agent Handled, connect it to the same Pivot Tables. Order Type = Online plus Agent = Adrian produces Online sales of 4700. Use the Multi-Select button or Ctrl-click to include Adrian and Roche: Online total becomes 9500. A numeric field such as Rating Given can also be inserted as a Slicer to select individual rating values.

Layout and clearing selections

1. Select a Slicer > Slicer ribbon > Columns. Set Order Type to 2 button columns, Agent to 3, and resize the floating objects to make horizontal controls.

2. Use the Clear Filter icon in each Slicer to restore full data. Clear all active Slicers if several are filtering the same report.

3. For every Pivot Table, right-click > PivotTable Options > Layout & Format; clear Autofit column widths on update; check Preserve cell formatting on update.

> Report Connections lists compatible Pivot Tables, generally those sharing the same cache or data model. Merely creating reports from similarly named tables does not guarantee a shared connection.

LESSON 26 | Timelines and interactive date filtering

Goal: select days, weeks, months, quarters and years while driving several Pivot Table reports.

How Timeline differs from Slicer

A Timeline is a horizontal interactive date control. Insert Slicer works with category fields, whereas PivotTable Analyze > Insert Timeline lists eligible date fields. The date column must contain real Excel dates, not text.

Create and test the Contact Date Timeline

1. Use a date-based Pivot Table; click inside it > PivotTable Analyze > Insert Timeline.
2. Select Contact Date, click OK, and position the Timeline above the dashboard.
3. Use the time-level dropdown to select Years, Quarters, Months or Days.
4. Click a month (such as Nov 2023), a quarter (Q4 2023), or a single day.
5. At the Days level, drag across a contiguous interval, such as 01-Nov through 07-Nov.
6. Select the Timeline border > Timeline ribbon > Report Connections to link other compatible Pivot Tables.

Expected values from the dated practice dataset

Selection	Expected sales
October 2023	6700
November 2023	25100
December 2023	4000
Q4 2023	35800
10-Nov-2023	2000
01-Nov to 07-Nov	7100
Online + November	14500

A Timeline may show date periods that have no matching orders; selecting such a period displays no matching Pivot Table records. This is not necessarily a fault. If both a Slicer and Timeline are active, they combine their restrictions. Clearing a Timeline does not clear a separate Order Type Slicer.

Date dashboard formatting

Place the Timeline above Overall Sales, Agent Sales and Product Sales reports; maintain reasonable spacing. For each Pivot Table, uncheck Autofit column widths on update and check Preserve cell formatting on update. Timeline options and styles vary by Excel version.

> In some earlier lesson examples the text of an example summary table did not match its transaction-level records. The monthly totals here are verified from the 16-row dated practice dataset, and sum to 35800.

LESSON 27 | End-to-end business performance dashboard

Goal: create the transcript's final combined report - 12 Pivot Tables covering overall, agent-wise and product-wise sales, satisfaction, orders and discounts - controlled by one Slicer.

Source schema and metrics

Prepare a source Excel Table named BusinessSalesData with columns: Order ID, Agent Handled, Product Name, Order Type, Final Price, Rating Given, Discount %. For the 15-row practice dataset, overall sales = 33700 and number of orders = 15.

Metric	Values field	Summarize by	Format
Overall Sales	Final Price	Sum	Currency, 0 decimals
Average CSAT	Rating Given	Average	Number, 1 decimal
Number of Orders	Order ID	Count	Number, 0 decimals
Average Discount	Discount %	Average	Percentage, 2 decimals

Build the 12 reports

Create the first Pivot Table in a clean dashboard worksheet, then copy the entire Pivot Table to make other compatible reports. Place reports with enough room for expansion. The three report groups are:

Group	Pivot Tables	Rows field
Overall business	Sales, Average CSAT, Orders, Average Discount	None
Agent-level	Sales, Average CSAT, Orders, Average Discount	Agent Handled
Product-level	Sales, Average CSAT, Orders, Average Discount	Product Name

Agent metrics from the 15-order source

Agent	Sales	Orders	Avg Rating	Avg Discount
Adrian	11000	5	7.6	7.40%
Albion	10500	5	7.2	8.60%
Roche	12200	5	8.0	9.00%

Product metrics from the 15-order source

Product	Sales	Orders	Avg Rating	Avg Discount
Cheese Pizza	7600	3	6.7	9.00%
Crispy Burger	6500	3	8.7	10.00%
Crispy Pizza	9400	4	8.3	7.50%
Paneer Pizza	6000	2	8.0	10.00%
Veg Burger	4200	3	6.3	6.00%

Add one Order Type Slicer across all reports

1. Select any Pivot Table > PivotTable Analyze > Insert Slicer > Order Type.
2. Select Slicer > Report Connections; check all 12 compatible Pivot Tables.
3. Click Online and verify overall sales of 17800 from 8 orders.
4. Click Physical Visit and verify sales of 15900 from 7 orders.
5. Clear the filter to restore 33700 sales and 15 orders.

Final dashboard finishing checklist

Use a title such as SALES PERFORMANCE ANALYSIS, place the Slicer above the reports, separate overall/agent/product sections, use friendly labels and consistent field-based number formats, and set Preserve cell formatting / disable Autofit on update for each Pivot Table. After adding source records, refresh all relevant reports. The average discount in this example is the simple mean of per-order percentages, not a weighted effective rate.

Business interpretation

Roche generated the highest agent revenue (12200), even though each agent handled five orders. Crispy Pizza led product revenue (9400); Crispy Burger had the highest average product rating (approximately 8.7). Sales, order count, satisfaction and discount should be considered together; one metric alone does not explain causes or profitability.

Master quick-reference and troubleshooting

Goal	Where to click / what to configure
Insert Pivot Table	Insert > PivotTable > From Table/Range
Change calculation	Right-click value > Value Field Settings
Number formatting	Value Field Settings > Number Format
Expand source	PivotTable Analyze > Change Data Source
Refresh current report	PivotTable Analyze > Refresh
Refresh workbook	Data > Refresh All
Sort revenue	Row dropdown > More Sort Options > Descending by sales
Label or Value filter	Row dropdown > Label Filters / Value Filters
Top 3 or Bottom 3	Row dropdown > Value Filters > Top 10
Group selected labels	Ctrl-click labels > right-click > Group
Group dates	Click a date > PivotTable Analyze > Group Field
Insert Slicer	PivotTable Analyze > Insert Slicer
Insert Timeline	PivotTable Analyze > Insert Timeline
Connect filters to reports	Select Slicer or Timeline > Report Connections
Protect widths	PivotTable Options > Layout & Format > disable Autofit

Errors and corrections to remember

If field names are missing, check the Pivot Table source. If an added row is ignored, switch the source to a growing Excel Table and refresh. If date grouping is unavailable, verify dates are genuine Excel dates without problematic blanks. If a Slicer does not control another report, inspect Report Connections and shared cache compatibility. If results differ from a lesson table, compare directly against the source rows, aggregation method, active filters and refresh state. If an item label changes in several Pivot Tables, remember shared caches can share metadata.

This reference handbook intentionally excludes practice assignments. It consolidates our worked lessons with step-by-step Excel navigation, datasets, calculations, outputs, common pitfalls and terminology.